Water Analysis

MACHEREY-NAGEL

Spectrophotometer *NANOCOLOR*® Advance



The future is spectral

- Universal – Large spectrum of evaluable test kits
- Intuitive – Icon-based menu navigation
- Safe – Automatic detection of interfering turbidity
- Robust – Certified outdoor usability

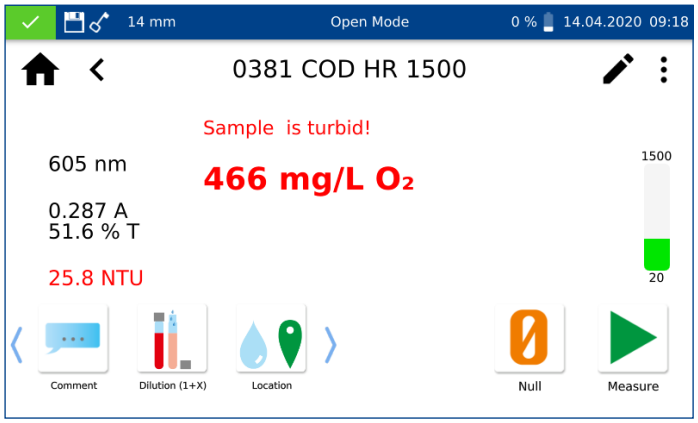
NANOCOLOR® Advance

Smart into the future

The spectrophotometer *NANOCOLOR*® Advance combines the most important features of our portable compact photometer PF-12^{Plus} and our high-precision spectrophotometer *NANOCOLOR*® *VIS* II. It provides precise and reliable measurement results in the daily laboratory routine and can also be used mobile due to its rechargeable battery. With its wavelength spectrum of 340–800 nm it allows the measurement of all photometrically evaluable test kits from MACHEREY-NAGEL. The *NANOCOLOR*® Advance can therefore be used universally in the entire spectrum of water and wastewater analysis.

Leave nothing to chance

The *NANOCOLOR*® Advance offers the user an automatic detection of interfering turbidity (NTU-Check). With an additional 860 nm LED, the device determines the nephelometric turbidity during each measurement of a tube test and warns the user of incorrect results. This option helps to avoid measuring errors and sets the basis for reliable measuring results.



Fast measurements – *NANOCOLOR*® Barcode Technology

- Simple and reliable routine analysis by fully automatic cuvette recognition via the integrated 2D barcode scanner
- The fast selection of the test method and the correct wavelength allows almost contactless operation

Easy user guidance – without extensive trainings

- The icon-based menu navigation based on smartphone and tablet ensures fun working with the device
- The intuitive handling avoids errors and ensures fast and reliable results
- The capacitive touch screen allows comfortable working and reliable operation even with gloves

Good to know

The *NANOCOLOR*® Advance has passed the vibration test according to Military Standard 810H only 514.8 CAT 4 and fulfills the requirements of protection class IP67. This underlines the robustness of the device and confirms its suitability for mobile analysis.

Turbidity – A source of error

Turbidity in the cuvette is often underestimated because it is not always visible. With the automatic turbidity control (NTU-Check), the turbidity is determined for each measurement and the user is warned in case of an interference.



A B A B

2 cuvettes (A | B) with the same COD concentration. The turbidity in cuvette B influences the result by up to 30 %.

Smart photometry

NANOCOLOR® Advance

Experience flexibility

The *NANOCOLOR*® Advance is the first photometer from MACHEREY-NAGEL with a 24 mm cuvette slot and thus extends the options for all areas of water analysis.

Besides the *NANOCOLOR*® tube tests and standard tests, the test kits from our *VISOCOLOR*® *ECO* and *VISOCOLOR*® Powder Pillow range can also be evaluated. The comprehensive and easy-to-understand menu for creating customer-specific methods makes the device an all-rounder in laboratory analysis and measurements on the road.

Concentrated scope of services – Versatile in use

- Evaluation of *NANOCOLOR*® tube tests, standard tests, *VISOCOLOR*® *ECO* test kits and *VISOCOLOR*® Powder Pillows
- Comfortable creation of customer-specific special methods
- Selection of the correct method in seconds via barcode, test number or intelligent search
- The robust design and the built-in battery allow full mobility without sacrificing precision and comfort
- The data export takes place directly via the integrated interfaces - prepared for future requirements

Comprehensive IQC options – Easy documentation and monitoring

- Versatile options for internal quality control allow monitoring of the device and the entire measuring system
- The test can be performed and documented by the user in a time-saving manner
- The integrated user administration enables complete documentation and individual adaptation to the user

Good to know

Universal cuvette slot

10, 20, 50 mm standard tests as well as 16 and 24 mm tube tests can be used with only two adapters.



Stable and with a large opening, the new 24 mm tube tests are ideally suited for the evaluation of our *VISOCOLOR*® Powder Pillows.

Good to know



All interfaces are located on the side and are therefore easily accessible. Switching on and off is conveniently done by a button below the display.



NANOCOLOR® Advance

Order information

Description	REF
Spectrophotometer *NANOCOLOR*® Advance incl. manual (quick start guide), protective cover, power supply unit with country specific adapters, USB stick, calibration cuvette, certificate and cleaning cloth in cardboard box	919750
Transport case for spectrophotometer *NANOCOLOR*® Advance	919757
NANOCONTROL NANOCHECK 2.0 Test solutions for checking photometric accuracy and linearity	925703
Colored reference standards to confirm device response (0–4 mg/L Cl_2) Contains four solutions that simulate test color at different concentrations, a certificate and instructions, in a rugged case	914820

Technical data

Specification	Description
Type:	Spectrophotometer with Reference Detector Technology (RDT)
Light source:	Halogen lamp
Optical system:	Single beam photometer with grating monochromator
Wavelength range:	340 nm–800 nm
Wavelength accuracy:	± 2 nm
Wavelength resolution:	1 nm
Wavelength calibration:	Automatically
Wavelength selection:	Automatically, Barcode, manually
Scan speed:	1 complete Scan < 3 min
Wavelength reproducibility:	+/- 0.1 nm
Spectral bandwidth:	< 4 nm
Photometric range:	+/- 3.0 A in the range 340–800 nm
Photometric accuracy:	0.003 A at 0.0–0.5 A; 1 % at 0.5–2.0 A
Photometric linearity:	< 0.5 % at 0.5–2.0 A; ≤ 1 % at > 2 A with neutral glass filters at 546 nm
Scattered light:	< 0.5 %
Measurement modes:	Over 200 pre-programmed tests and special methods; 99 fully programmable methods; absorbance; transmission; factor; kinetics; scan; nephelometric turbidity measurement
Turbidity measurement:	Nephelometric turbidity measurement (16 mm and 24 mm) 1–1000 NTU
Cuvette slot:	Tube test 16 mm and 24 mm OD; Standard test 10 mm, 20 mm, 40 mm and 50 mm
Data memory:	1000 Measured values, 1000 IQC results, 100 spectra; GLP-conform
Display:	Backlit colored 5″ display with touch screen
Operation:	Barcode technology; icon-based display menu navigation; capacitive touch screen
Languages:	DE / EN / FR / ES / NL / IT / HU / PL / CZ / PT-Br / TK / BG / RO / DK
External light:	Insensitive; open cuvette slot
Interfaces:	LAN (CAT 6; only use shielded cables with a maximum length of 20 m) 2 × USB (Host), 1× USB (Function)
Protection class:	IP67
Update:	Via USB stick
Dimensions:	325 × 280 × 130 mm (L x W x H)



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KATEN100226 Flyer *NANOCOLOR*® Advance en2 / x / xx / 06.2021

Management System EN ISO 13485:2016 ISO 9001:2015
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